



Targeted and untargeted mass spectrometry-based proteome investigation of urine in Parkinson's disease

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Summary (or Abstract)

We analyzed 1,164 urine samples from the PPMI PD cohort including carriers of a pathogenic genetic variant and healthy controls. All samples were subjected to LC-MS/MS analysis to determine urinary proteome profiles using a data-independent acquisition strategy. Furthermore, we performed a targeted analysis of Rab10 and LRRK2 using stable isotope-labeled reference peptides. This enabled us to quantify the Rab10-T73 phosphorylation site and to determine the Rab10 phosphorylation stoichiometry in most samples. The LRRK2 phosphorylation stoichiometry could be determined only in a subset of the samples due to the low abundance of phosphopeptides in urine.

Method

Proteins were extracted from neat urine samples and digested into peptides by using the MStern blotting sample preparation protocol (Berger et al. 2015). The protocol was slightly modified and semi-automated on an Agilent Bravo liquid handling system. To determine urinary proteome profiles, purified peptides were loaded on Evosep Evotips, separated via an online-coupled Evosep One HPLC (Bache et al. 2018) and analyzed on a Bruker timsTOF Pro mass spectrometer with a data-independent acquisition method (Meier et al. 2020) and a gradient length of 44 minutes. Targeted analysis was performed using the same peptide extract but using an online-coupled EASY-nLC1200 HPLC with a gradient length of 21 minutes. Analysis of raw spectra was performed with DIA-NN (discovery data) (Demichev et al. 2020) and Skyline (targeted data) (Pino et al. 2017).

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